Mean, Median, and Baselines

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

One constant-baseline experiment record.

Trace fitted values,evaluation errors and a documented training-value change.

Keep one habit: W1–W3/W5 → selected new targets → numeric project reference → changed-case return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Regression and Residuals. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Precision and Recall. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Scores and Calibration Intuition. Give an example and a mistake that the idea helps you avoid.

Regression and Residuals

> **My remembered idea, example and correction**

Precision and Recall

> **My remembered idea, example and correction**

Scores and Calibration Intuition

> **My remembered idea, example and correction**

W1 · Explain the mechanism

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main training[2,4,6,12]cm gives mean6and median5;validation[4,8]gives errors[2,2]and[1,3],both MAE2cm. Changing training12to40gives mean13,median5;fixed validation errors[9,5]MAE7versus[1,3]MAE2. Picture training[2,4,6,20]gives mean8,median5;eval[4,10]gives both MAE3. Pictured20→8makes both constants5,still evalMAE3. Mean=sum/count;median is middle sorted value or average of the two middles for an even count.

  Evaluation never fits the constant.

- Reference for this exercise: open this week's student lesson, "Mean, Median, and Baselines", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- Find the **mean** and **median** of the worked **training** targets.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Trace the current experiment

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main training[2,4,6,12]cm gives mean6and median5;validation[4,8]gives errors[2,2]and[1,3],both MAE2cm. Changing training12to40gives mean13,median5;fixed validation errors[9,5]MAE7versus[1,3]MAE2. Picture training[2,4,6,20]gives mean8,median5;eval[4,10]gives both MAE3. Pictured20→8makes both constants5,still evalMAE3. Mean=sum/count;median is middle sorted value or average of the two middles for an even count.

  Evaluation never fits the constant.

- Calculate MAE for both baselines on [4, 8].

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Repair the reasoning

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main training[2,4,6,12]cm gives mean6and median5;validation[4,8]gives errors[2,2]and[1,3],both MAE2cm. Changing training12to40gives mean13,median5;fixed validation errors[9,5]MAE7versus[1,3]MAE2. Picture training[2,4,6,20]gives mean8,median5;eval[4,10]gives both MAE3. Pictured20→8makes both constants5,still evalMAE3. Mean=sum/count;median is middle sorted value or average of the two middles for an even count.

  Evaluation never fits the constant.

- Replace 12 by 40. What changes?

> **My reasoning and evidence**

MAIN LESSON · SECTION 16

W5 · Plan the next check

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main training[2,4,6,12]cm gives mean6and median5;validation[4,8]gives errors[2,2]and[1,3],both MAE2cm. Changing training12to40gives mean13,median5;fixed validation errors[9,5]MAE7versus[1,3]MAE2. Picture training[2,4,6,20]gives mean8,median5;eval[4,10]gives both MAE3. Pictured20→8makes both constants5,still evalMAE3. Mean=sum/count;median is middle sorted value or average of the two middles for an even count.

  Evaluation never fits the constant.

- Why not predict the evaluation **mean** as the baseline?

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected experiment task

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Use the selected training values1,3,5,15to fit both constants,then keep them fixed while scoring the supplied3,9targets. All values use the same numeric target/unit for this exercise. Neither the pictured training20nor the main training12belongs in this task. No final-test population or reliability claim is supplied.

- **Training** [1, 3, 5, 15], evaluation [3, 9]. Fit both constants and calculate MAE.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 9

Project · Keep the experiment evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Document any numeric baseline used in Plant Classifier feature checks, including unit, distribution, and **outlier** decision.

- The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

MAIN LESSON · SECTION 10

Finish · Judge your evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- - I can explain the mechanism.

- I can show evidence.
- I can handle a changed case.
- I can name a limit.

  Mark each as not yet, with help or independently, and attach supporting work.

- Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Use W2: Score separately for Constant **prediction** baselines. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Use the worked case for Constant **prediction** baselines. Proposed explanation: "A final answer is enough". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier learning target

REFERENCE · SECTION 5

Reference · Earlier learning target

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- For **training** targets [1,3,11], compute **mean** and **median constant baseline**s. Compare total absolute **training** error for each. Where must final performance be measured?

▼ **Extra practice · Earlier complete case**

REFERENCE · SECTION 6

Extra practice · Earlier complete case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- **Training** heights are 2,4,12 cm. Compare constant **mean** and **median** baselines.
- Compute both and **training absolute error**s.
- Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ Extra practice · Changed case

REFERENCE · SECTION 7

Extra practice · Changed case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- For **training** heights 2, 3 and 10 cm, calculate **mean** and **median**. Which constant is less influenced by the high value?
- My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 20 Test and reflect

▼ Extra practice · Design a revealing test

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Extra practice · Earlier transfer**

REFERENCE · SECTION 11

Extra practice · Earlier transfer

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- For **training** targets [1,3,11], compute **mean** and **median constant baseline**s. Compare total absolute **training** error for each. Where must final performance be measured?

- Record support used: none, hint or teacher explanation.

My independently reasoned answer

▼ **Reference · Earlier W4 record**

REFERENCE · SECTION 15

Reference · Earlier W4 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- **Training** [1, 3, 5, 15], evaluation [3, 9]. Fit both constants and calculate MAE.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 18

Extra practice · Pictured case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Fit two constant height baselines and compare them on held-out examples.
- **Training** heights [2, 4, 6, 20] cm. Evaluation heights [4, 10] cm. Fit constants only from **training**.
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.
- Does either original baseline win under evaluation MAE? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.
- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

AI ACADEMY / AI-4/20

Mean, Median, and Baselines

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Fit mean

Training mean=(2+4+6+20)/4=8 cm.

↓

2. Fit median

Training median=(4+6)/2=5 cm.

↓

3. Evaluate constants

Mean MAE=(4+2)/2=3 cm; median MAE=(1+5)/2=3 cm.

CHANGE ONE THING

Replace training 20 with 8 while keeping evaluation unchanged.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 19

Reference · Project portfolio milestone

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Project: Responsible Plant Classifier
- Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.
- For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 20

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Fit two constant height baselines and compare them on held-out examples.
- Which labelled step matches this detail? **Mean** MAE=(4+2)/2=3 cm; **median** MAE=(1+5)/2=3 cm.
- 1. Fit **mean**
- 2. Fit **median**
- 3. Evaluate constants
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Evaluate constants **Mean** MAE=(4+2)/2=3 cm; **median** MAE=(1+5)/2=3 cm. Read the picture in order. **Mean** MAE=(4+2)/2=3 cm; **median** MAE=(1+5)/2=3 cm.

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: Replace **training** 20 with 8 while keeping evaluation unchanged. Which conclusion is supported by the supplied facts?
- 1. Both fitted constants become5 cm and both evaluation MAEs remain3 cm.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: Both fitted constants become5 cm and both evaluation MAEs remain3 cm. **Mean** is now20/4=5; **median** stays5. The earlier **training outlier** affected the **mean**.

Repair and keep the record

Review Regression and Residuals (ai-4/19). Goal: Distinguish numeric **prediction** from classification. Short practice: Which statistic reveals the size of errors without cancellation?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Precision and Recall (ai-4/17). Goal: Choose the denominator that answers the question. Short practice: Were the original precision and recall denominators the same?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Scores and Calibration Intuition (ai-4/14). Goal: Interpret calibration using groups of forecasts. Short practice: Does a calibrated 0.8 score **mean** every such leaf is spotted?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

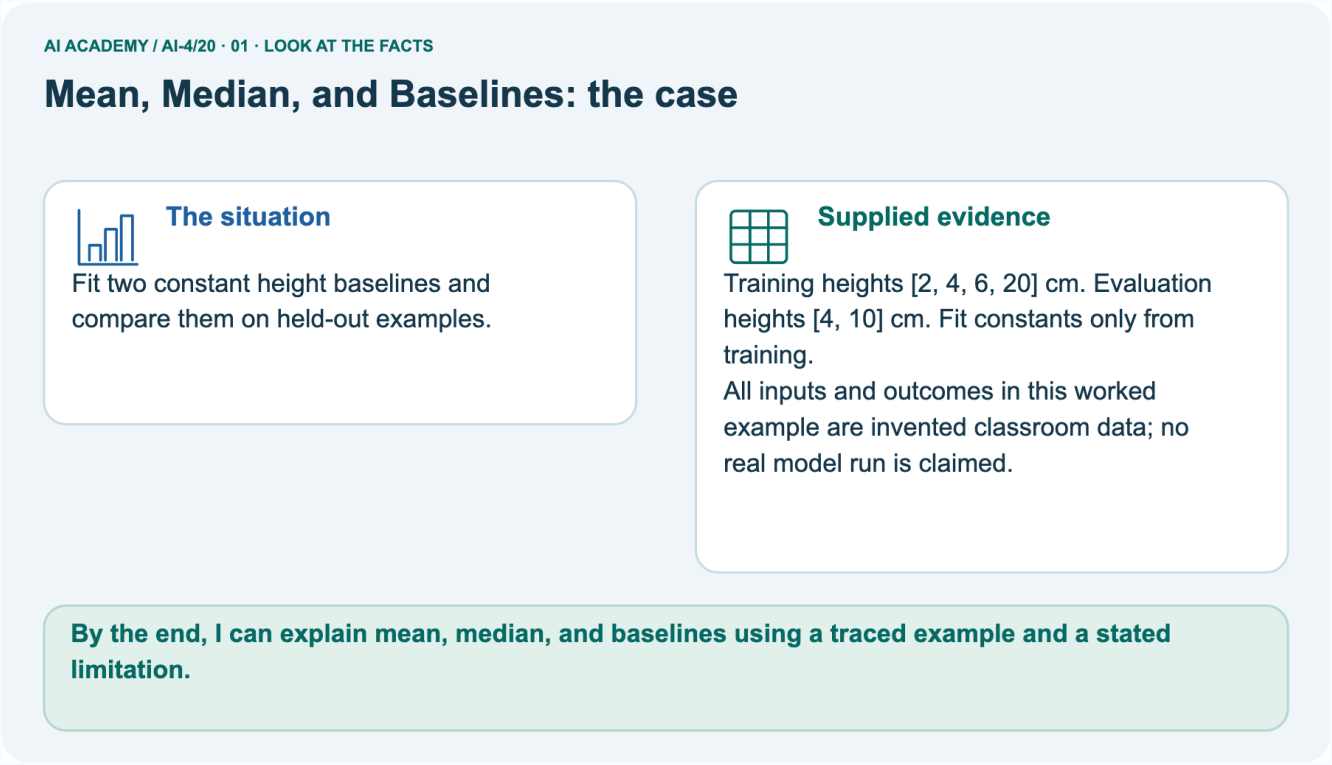
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 21

Reference · Six-picture response sheets

Choose the relevant picture instead of repeating every task.

01. Mean, Median, and Baselines: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

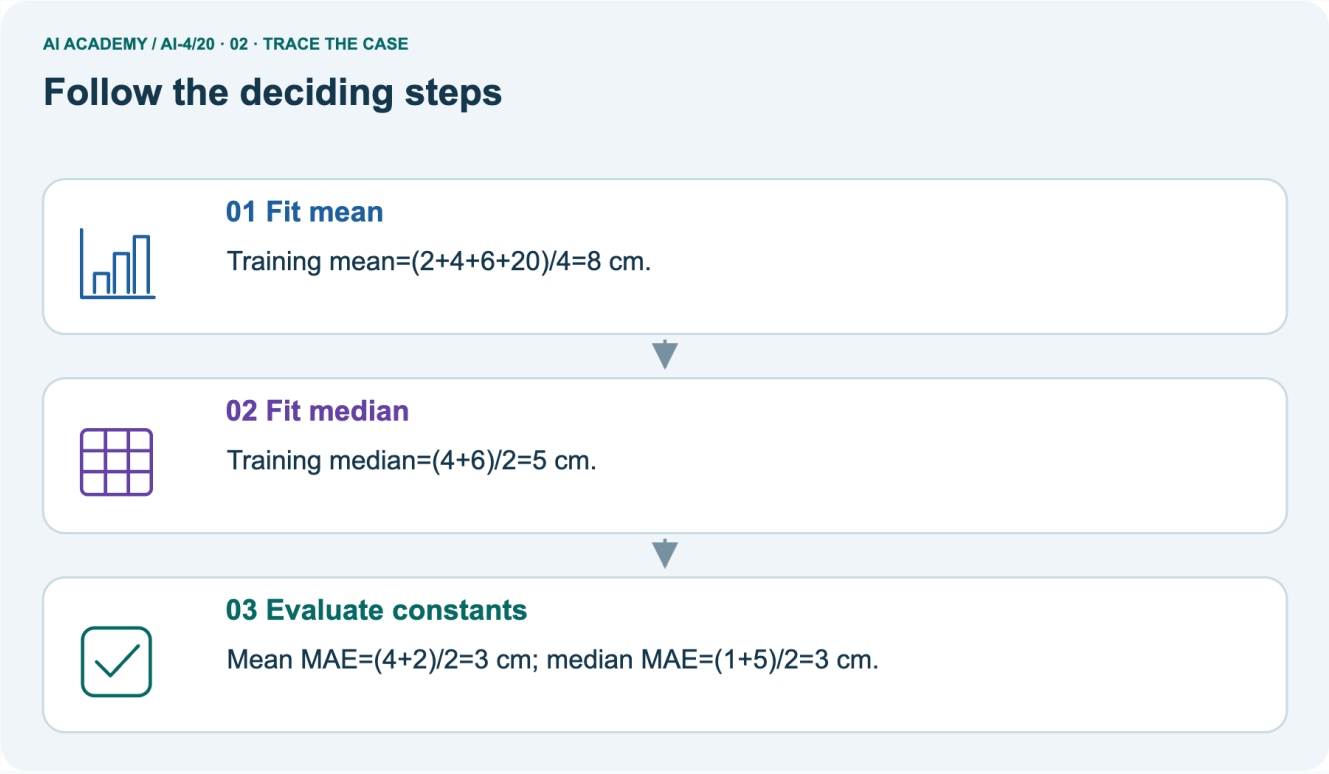
Fit two constant height baselines and compare them on held-out examples. Supplied facts: **Training** heights [2, 4, 6, 20] cm. Evaluation heights [4, 10] cm. Fit constants only from **training**.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Fit **mean**: **Training mean**=(2+4+6+20)/4=8 cm. → Fit **median**: **Training median**=(4+6)/2=5 cm. → Evaluate constants: **Mean** MAE=(4+2)/2=3 cm; **median** MAE=(1+5)/2=3 cm.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Constant prediction baselines at a glance

AI ACADEMY / AI-4/20 · 03 · INSPECT THE LESSON MODEL

Constant prediction baselines at a glance

Part	Evidence or action	Interpretation
Training targets	Fit mean 6	Freeze constant
Training targets	Fit median 5	Freeze constant
Evaluation targets	Compare fixed predictions	Both MAE 2 here

Mean and median answer different training objectives. A mean minimizes total squared error among constants; a median minimizes total absolute error. This does not guarantee either wins on a new evaluation set. With extreme values, the mean can move much more than the median.

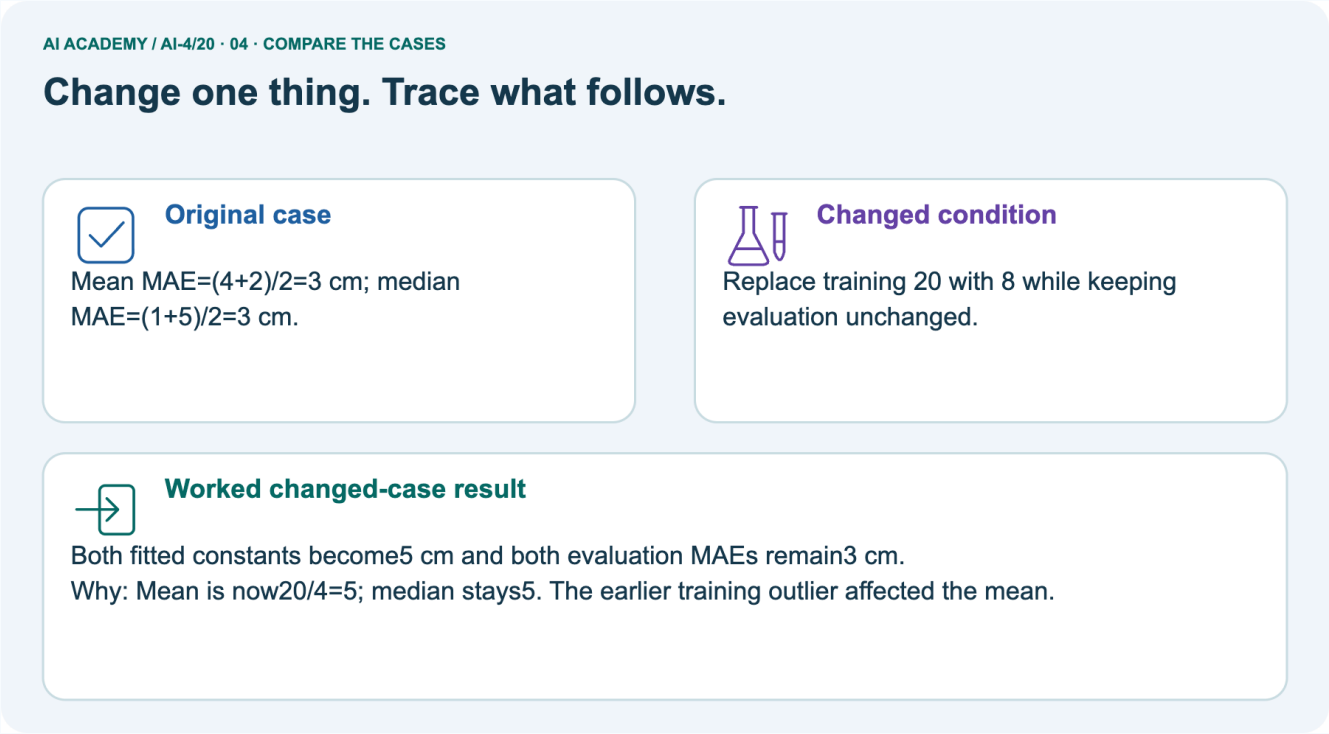
Follow the labelled picture with the matching explanation. Open the vector diagram

Constant **prediction** baselines at a glance. **Training** targets | Fit **mean** 6 | Freeze constant. **Training** targets | Fit **median** 5 | Freeze constant. Evaluation targets | Compare fixed **prediction**s | Both MAE 2 here

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

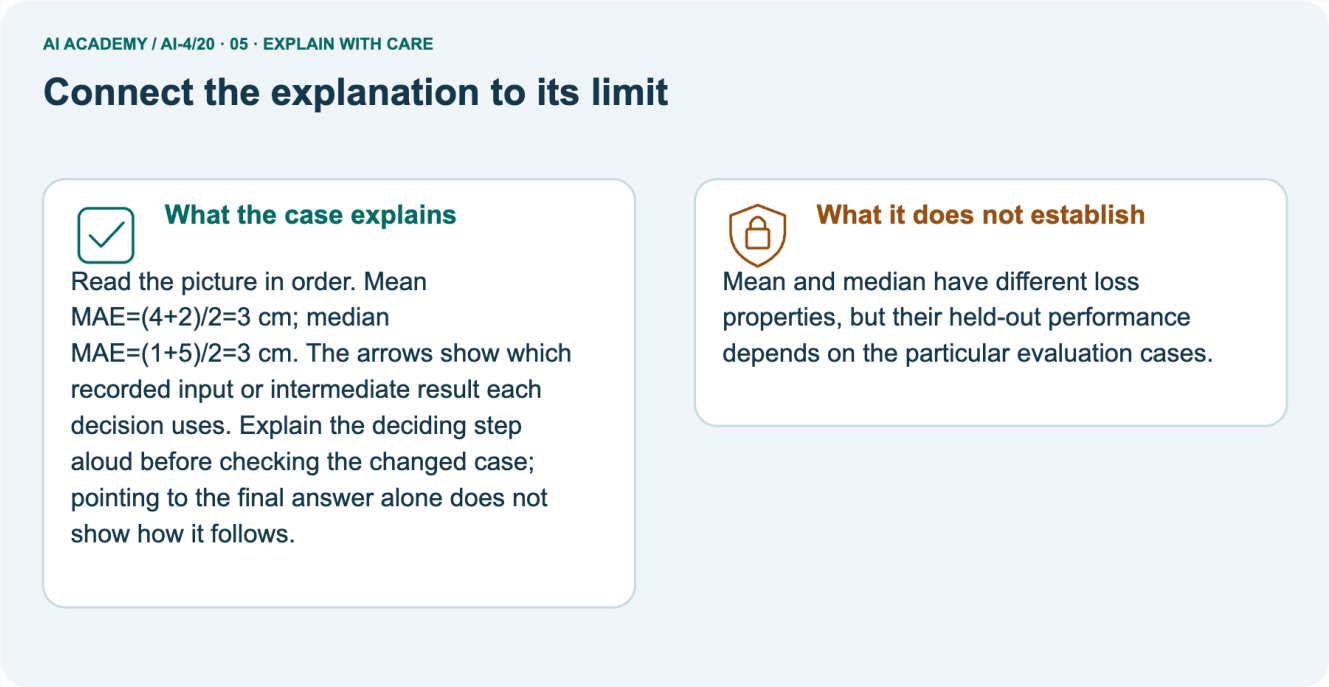
Original case: **Mean** MAE=(4+2)/2=3 cm; **median** MAE=(1+5)/2=3 cm. Changed condition: Replace **training** 20 with 8 while keeping evaluation unchanged.

Expected result: Both fitted constants become5 cm and both evaluation MAEs remain3 cm. Reason: **Mean** is now20/4=5; **median** stays5. The earlier **training outlier** affected the **mean**.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. **Mean** MAE=(4+2)/2=3 cm; **median** MAE=(1+5)/2=3 cm. The arrows show which recorded **input** or intermediate result each decision uses.

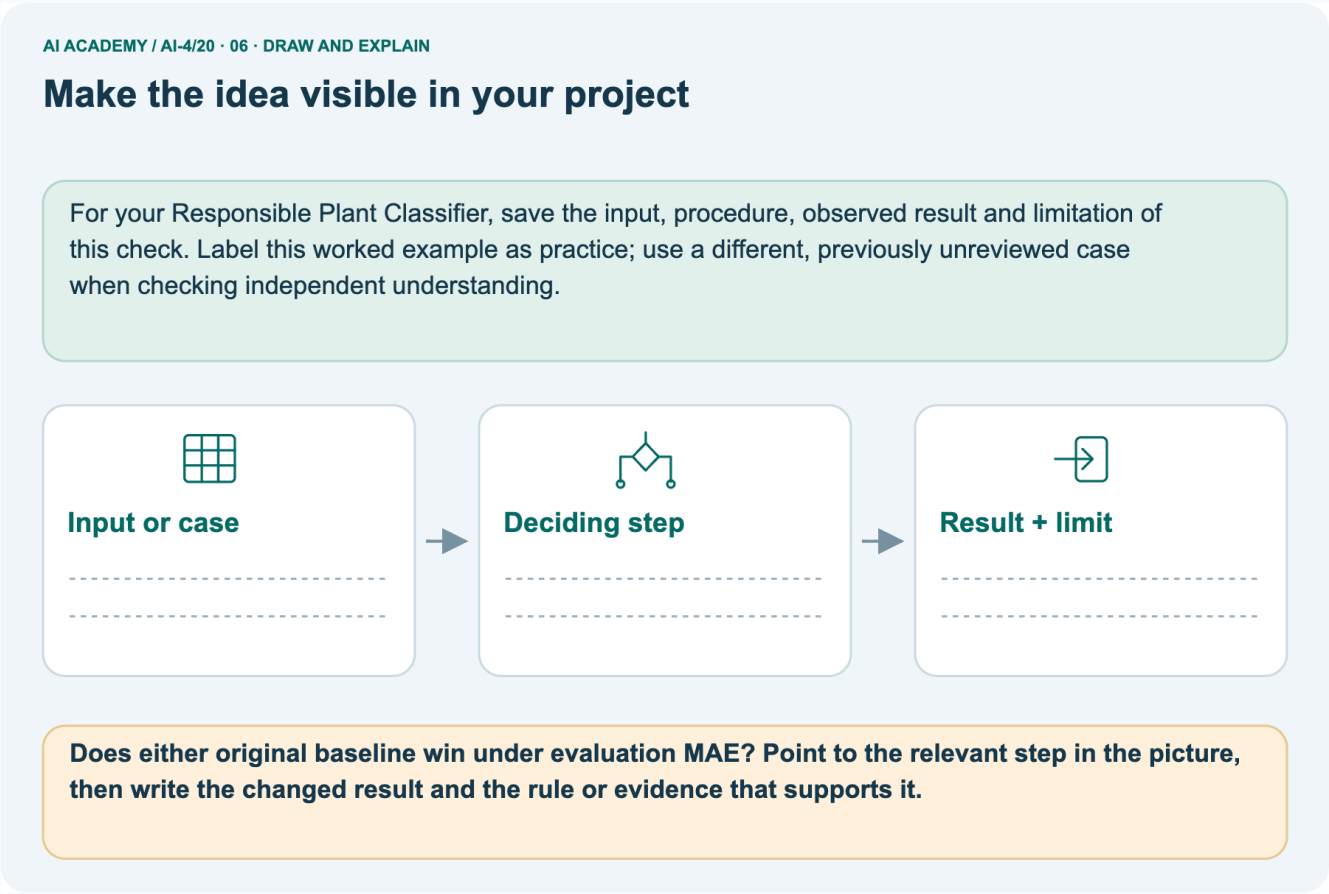
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: **Mean** and **median** have different loss properties, but their held-out performance depends on the particular evaluation cases.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Does either original baseline win under evaluation MAE? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Mean, Median, and Baselines: the case

02. Follow the deciding steps

03. Constant prediction baselines at a glance

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.